

Influence of Ethanol on the Solubility, Ionization and Permeation Characteristics of Ibuprofen in Silicone and Human Skin

R.M. Watkinson^a C. Herkenne^b R.H. Guy^{b, c} J. Hadgraft^d G. Oliveira^d
M.E. Lane^d

^aVictorian College of Pharmacy, Monash University, Parkville, Vic., Australia; ^bDebio Recherche Pharmaceutique SA, Martigny, Switzerland; ^cDepartment of Pharmacy and Pharmacology, University of Bath, Bath, and ^dDepartment of Pharmaceutics, The School of Pharmacy, University of London, London, UK

Key Words

Ibuprofen · Ethanol · pK_a · Silicone · Skin

Abstract

In the context of topical and transdermal delivery, cosolvents may be expected to enhance solubility, modify drug partitioning behaviour and alter skin barrier properties. In the present study, we have investigated systematically the influence of increasing amounts of the cosolvent ethanol on the solubility, ionization, and permeability characteristics of ibuprofen in silicone and in skin. The possibility of ibuprofen self-association was also examined. As the amount of ethanol was raised from 0 to 100%, ibuprofen solubility was enhanced 5,500-fold relative to its aqueous solubility. Up to 50%, the pK_a of ibuprofen shifted from 4.44 to 5.68. Infrared spectroscopy confirmed the presence of ibuprofen in both monomer and dimer forms in vehicles containing 75 and 100% ethanol. In silicone membranes, the flux of ibuprofen increased 8-fold over the range of ethanol concentrations studied (0–100%) relative to aqueous solutions, with the highest flux observed for 100% ethanol. However, in skin, the flux of ibuprofen was optimal for 50:50 and 75:25 ethanol-water vehicles (>10-fold flux enhancement). The lower ibuprofen flux from pure ethanol may reflect the alcohol's

ability to dehydrate human skin. In general, the flux data confirm that increasing ethanol content enhances the solubility of ibuprofen in silicone and human skin as well as in the vehicle itself.

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Introduction

To understand the role of formulation components in topical drug delivery, it is necessary to separate the various contributing factors to the permeation process which may include drug thermodynamic activity, the ionization state of the permeant and any vehicle effects on membrane properties. For ionizable permeants there is also the possibility that the ionized species will permeate, even though its partitioning into the stratum corneum may be orders of magnitude lower than that for the unionized moiety [1].

The steady-state flux of a drug through the stratum corneum is described by Fick's first law of diffusion (equation 1)

$$J = \frac{K \cdot D}{h} \cdot C_{\text{vehicle}} \quad (1)$$

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Majella E. Lane
Department of Pharmaceutics, The School of Pharmacy
University of London
London WC1N 1AX (UK)
Tel. +44 207 753 5821, Fax +44 870 165 9275, E-Mail majella.lane@btinternet.com

where J represents drug flux, D is the drug diffusion coefficient, K is the partition coefficient for drug between the stratum corneum and the vehicle, C_{vehicle} is the drug concentration in the vehicle and h is the diffusional path length for the drug.

If a cosolvent does not affect membrane barrier properties, its inclusion in the formulation of a weakly acidic drug of the form HX is expected to

- (1) increase the unionized fraction (HX) of the drug. Since the latter has a higher lipophilicity relative to X^- , the overall skin flux, which is proportional to the skin-formulation partition coefficient, K (equation 1) should theoretically be enhanced;
- (2) increase drug solubility in the formulation. If the drug concentration is kept constant, this effectively decreases the degree of saturation and hence should lead to a lower flux. The precise outcome will depend ultimately on the nature of the cosolvent, the pH of the formulation and the pK_a of the drug.

Many non-steroidal anti-inflammatory drugs are organic acids with pK_a s ranging from 3 to 5. At the pH of the skin surface (~ 5), such compounds will be ionized. In the deeper layers of the skin, a less acidic environment is encountered and the degree of ionization of such molecules will therefore increase with consequent effects on solubility and permeation. Ibuprofen is one of the most widely available non-steroidal anti-inflammatory drugs for the management of inflammation and is available in both gel and cream formulations. Although the permeation of ibuprofen has been studied from a range of vehicles [2], there has been, to our knowledge, no systematic investigation into the manner in which formulation components such as cosolvents may influence the ionization and permeability of ibuprofen into and through the skin. The aim of the present study therefore was to examine the influence of increasing amounts of the cosolvent ethanol on the solubility, ionization and permeation characteristics of ibuprofen through silicone and human skin. As saturated solutions were used throughout this work, a further aim was to investigate the possibility of self-association of ibuprofen in the different cosolvent mixtures.

Materials and Methods

Materials

Ibuprofen was a generous gift from Wyeth Consumer Healthcare, Havant, UK. All other materials were of reagent grade and were obtained from VWR, UK.

Solubility of Ibuprofen in Aqueous-Ethanol Mixtures

Excess drug was added to each solvent or cosolvent mixture and stirred with a magnetic bar for 48 h in a water bath maintained at 32°C. Solutions were then centrifuged for 15 min at 4,000 g and the supernatant was diluted and assayed either by HPLC or UV spectroscopy. For HPLC analysis, the sample was diluted in isopropanol, followed by appropriate dilution in methanol. The stationary phase was an Apex reverse-phase ODS 5- μ m packed column (250 $\times$ 4.6 mm) and a mobile phase of 65% acetonitrile and 35% phosphate buffer (adjusted to pH 3.2 using orthophosphoric acid). UV detection was employed at 225 nm and a flow rate of 1.2 ml·min⁻¹. The retention time was approximately 5 min, and calibration curves were constructed on the basis of peak area measurements using >5 standard solutions. Experiments were performed in triplicate and mean values with standard deviation and coefficient of variation were calculated. Reproducibility was evaluated prior to injection of the samples and during analysis, and CV values were <5% in all experiments.

Saturated Aqueous Solubility as a Function of pH

Excess drug was added to 10 ml of phosphate-citrate buffer [3] at different pH values and stirred with a magnetic bar for 48 h in a water bath maintained at 32°C. The ionic strength was controlled by inclusion of 3 M NaClO₄ in the buffer solutions. Solutions were centrifuged for 30 min at 10,000 rpm. The supernatant was diluted and assayed by UV spectroscopy. Experiments were performed in triplicate and mean values with SD and CV were calculated.

pK_a and log P and log D Determination

The pK_a values of ibuprofen were determined using published methods [4, 5]. Ibuprofen was titrated in mixtures of water and ethanol using a GlpKa apparatus (Sirius Analytical Instruments, Forrest Row, UK). The ionic strengths of the solutions were set at 0.15 M with KCl. Yasuda-Shedlovsky plots were used to calculate 'aqueous' pK_a values [6, 7]. Using octanol as the partition solvent, log P values were also determined by potentiometric titration. The pK_a log P program (version 5.1, ACD Ltd.) was also used to predict the distribution coefficients (log D) which were subsequently compared with the experimentally derived values.

Permeation Studies in Silicone Membrane and Skin

The silicone membranes used in this work were from Samco (Nuneaton, UK) with a measured thickness of 400 μ m as described previously [8]. Full-thickness human skin from cosmetic surgery, obtained following informed consent and with appropriate institutional ethical approval was used for skin permeation studies. The epidermis was separated by immersing the skin in water maintained at 60°C for 1 min [9]. The tissue was then pinned to a corkboard and the epidermis was very carefully peeled away from the dermis. The epidermis was immersed in cold water, mounted onto filter paper, and stored in a freezer at -20°C until required. Prior to the diffusion experiment the skin was defrosted and cut to appropriate size.

Permeation experiments ($n \geq 4$) were conducted in Franz diffusion cells with a receptor volume of 2.5 ml, and a diffusion area of 1 cm². For all permeation experiments the receptor fluid was pH 7.4 phosphate-buffered saline. The diffusion experiments were performed under occluded conditions by sealing the donor compartment with microscope coverslips. Studies were also con-

ducted with membranes pretreated with ethanol for 24 h to further investigate the influence of ethanol on ibuprofen permeation.

Flux values were calculated from the cumulative amount of drug diffusing into the receptor compartment as a function of time. The slope of this profile was determined once steady state had been reached. Values of the permeability coefficient k_p , Kh and D/h^2 were calculated by fitting the data to equation 2:

$$Q = c_v K h \left[\frac{D}{h^2} t - \frac{1}{6} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{h^2} \exp \left(n^2 \pi^2 \frac{D}{h^2} t \right) \right] \quad (2)$$

where Q is the cumulative mass of permeant that passes through a unit area of a membrane in a time t [10]. Differences between parameters were evaluated using ANOVA with a significance of $p \leq 0.05$.

Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy

The presence of ibuprofen monomer and dimer species was evaluated using multibounce attenuated total reflectance Fourier transform infrared (ATR-FTIR). A Bruker Tensor 27 instrument fitted with a Specac multiple reflection accessory with a zinc selenide crystal was used and data were acquired using OPUS software, version 5.5 (Bruker, Germany).

Results and Discussion

Solubility Studies

Addition of 25% ethanol increased the solubility of ibuprofen to 343 mg/ml, a 3,800-fold enhancement relative to the drug's aqueous solubility (table 1). In pure ethanol, the solubility is 5,500-fold higher than in water. Ibuprofen has a reported pK_a of 4.41 [11], explaining the considerable solubility enhancement (50-fold) over the pH range 2–6 (table 2). Similar solubility data have been reported over a comparable pH range elsewhere [12].

pK_a Determination

The pK_a of ibuprofen in water, and in 25:75 and 50:50 ethanol-water mixtures, was determined by potentiometry. From Yasuda-Shedlovsky plots constructed for ibuprofen in ethanol the relative fraction of drug which is ionized can be determined as a function of pH. Increasing the ethanol content from 0 to 50% shifted the pK_a of ibuprofen from 4.44 to 5.68 (fig. 1). In water, approximately 80% of ibuprofen is ionized at pH 5, which is close to that typically found at the skin surface. However, in 50:50 (v/v) ethanol-water, the pK_a shift reduces the percentage of the ionized form of ibuprofen to ~15% at pH 5. Therefore, a greater amount of unionized ibuprofen is available with increasing ethanol content.

Table 1. Solubility of ibuprofen in ethanol and water vehicles (n = 3)

Ethanol in vehicle (v/v), %	Solubility, mg/ml
0	0.09 ± 0.06
25	343 ± 6.5
50	453 ± 21
75	489 ± 20
100	500 ± 4.1

Table 2. Aqueous solubility of ibuprofen as a function of pH (n = 3)

pH	Solubility, mg/ml
2.0	$2.8 \times 10^{-2} \pm 0.4 \times 10^{-2}$
3.0	$3.1 \times 10^{-2} \pm 0.03 \times 10^{-2}$
5.0	0.41 ± 0.05
6.0	1.33 ± 0.04

Log D

There is a good correlation between the measured and predicted log D values as shown in figure 2; the results also agree well with previous measurements [1]. The log D of unionized ibuprofen is 4.0, whereas that of the ionized form is 0.08. At the pH of the skin surface (~5), log D is predicted to be 3.2. With increasing amounts of ethanol, log D increases further because of the increase in pK_a and the suppression of ibuprofen ionization.

ATR-FTIR Spectroscopy

The possibility of self-association was investigated by ATR-FTIR as such behaviour is typical of carboxylic acids in organic media [13]. The infrared reference spectrum of ibuprofen exhibits a sharp C = O bond absorbance peak around $1,710 \text{ cm}^{-1}$ which can be used to monitor the presence of monomer or dimer in solution. In order to form dimers, the ibuprofen molecules will associate through interactions between the hydroxyl hydrogen and the carbonyl oxygen of two adjacent molecules. This interaction will introduce an extra constraint to the vibration of the C = O bond, therefore decreasing its stretching frequency. Consequently, the ibuprofen dimer will absorb at a lower wavenumber relative to the monomer [14]. The results obtained with saturated solutions of ibuprofen in 25, 50, 75 and 100% (v/v) ethanol in water are shown in figure 3. No C = O absorbance peak of ibuprofen was seen because of significant water interference in

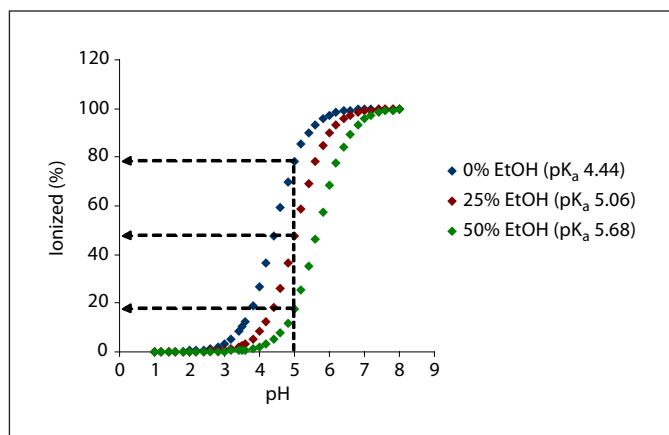


Fig. 1. Modulation of ibuprofen pK_a by increasing the percentage of cosolvent (ethanol) in the formulation.

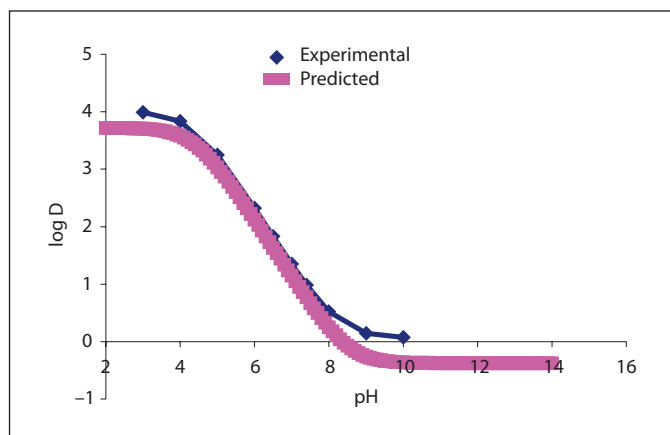


Fig. 2. Comparison of experimental and predicted log D values of ibuprofen.

vehicles with higher water content [25 and 50% (v/v) ethanol in water] and in which ibuprofen solubility is lower. However, it was possible to detect the ibuprofen C = O absorption in 75:25 (v/v) ethanol-water and in pure ethanol. The C = O peak occurred at a lower frequency (arrow a), consistent with the presence of ibuprofen dimers. There was also a C = O signal at a higher frequency in pure ethanol (arrow b), indicating that some ibuprofen also exists in the monomer form.

Permeation Studies in Silicone Membrane and Skin

Ibuprofen fluxes and permeability coefficients through silicone membranes are presented in table 3. Addition of 25% ethanol induced a 2-fold increase in flux. Increasing the amount of ethanol to 50% again doubled the flux. While 75:25 (v/v) ethanol-water did not cause a further significant increase in transport, the use of pure ethanol resulted in the highest flux observed, $\sim 1.5 \text{ mg/cm}^2/\text{h}$.

The slope of the Yasuda-Shedlovsky plot used to calculate the aqueous pK_a is inversely proportional to the average ionic diameter of the solvated molecule [7]. From the plots constructed in this work, it appears that there is a change in the solvation structure at around 30:70 (v/v) ethanol-water, as there is a change in slope at this cosolvent mixture (data not shown). It is possible that the enhanced flux reflects the faster permeation of ethanol relative to water and that ibuprofen within an ethanol 'cage' is taken across the membrane. Because of the change in pK_a with solvent polarity, and the greater amount of unionized ibuprofen available for permeation, the partitioning of ibuprofen into the membrane should be affected.

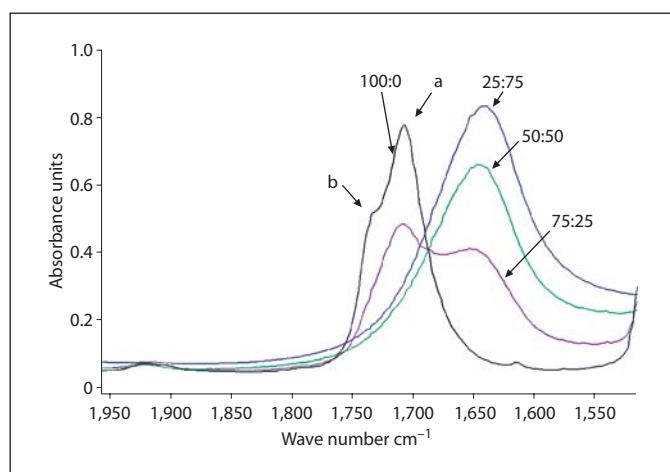


Fig. 3. ATR-FTIR spectra of ibuprofen in 25:75, 50:50, 75:25 and 100:0 (%v/v) ethanol-water solutions.

Analysis of the data confirmed that the Kh parameter decreased for ethanol levels above 50%. Reduction in the silicone membrane:vehicle partition coefficient with increasing ethanol content for oestradiol permeation from ethanol-water systems has also been observed [15]. The diffusion parameter increased when ethanol was present, although the values for water alone and for pure ethanol were not significantly different ($p < 0.05$). As ethanol may interact with the membrane, either to improve drug solubility, or by another mechanism, further studies were conducted in which the membrane was pretreated with ethanol.

Table 3. Steady-state fluxes, thermodynamic and kinetic parameters, and permeability coefficients for the permeation of ibuprofen from saturated formulations containing ethanol and water through silicone membranes (n = 4)

EtOH-water, %v/v	J, $\mu\text{g cm}^{-2} \text{h}^{-1}$	K·h, μm	D/h ² , h ⁻¹	k _p , cm h ⁻¹
0:100	186 ± 11	1.8 ± 0.2	1,368 ± 684	2.09 ± 0.14
25:75	420 ± 2.5*	1.3 ± 0.85	1.12 ± 0.50	1.22 × 10 ⁻³ ± 6.84 × 10 ⁻⁵
50:50	916 ± 79*	0.60 ± 0.44	3.38 ± 1.01	2.02 × 10 ⁻³ ± 2.45 × 10 ⁻⁴
75:25	945 ± 61*	0.47 ± 0.24	4.68 ± 2.23	1.91 × 10 ⁻³ ± 5.04 × 10 ⁻⁵
100:0	1,495 ± 116*	5.6 × 10 ⁻³ ± 2.0 × 10 ⁻³	612 ± 288	2.90 × 10 ⁻³ ± 3.62 × 10 ⁻⁴

* p < 0.05 significantly different from flux values from aqueous vehicle.

Table 4. Steady-state fluxes, thermodynamic and kinetic parameters, and permeability coefficients for the permeation of ibuprofen from saturated formulations containing ethanol and water through silicone membranes pretreated with ethanol (n = 4)

EtOH-water, %v/v	J, $\mu\text{g cm}^{-2} \text{h}^{-1}$	K·h, μm	D/h ² , h ⁻¹	k _p , cm h ⁻¹
25:75	449 ± 31.3	0.26 ± 0.04	39.6 ± 25.2	9.43 × 10 ⁻⁴ ± 8.46 × 10 ⁻⁵
75:25	712 ± 29.0*	8.48 × 10 ⁻³ ± 1.44 × 10 ⁻³	1,656 ± 324	1.37 × 10 ⁻³ ± 0.42 × 10 ⁻³
100:0	1,681 ± 80*	7.93 ± 3.35	4.97 ± 2.03	3.42 × 10 ⁻³ ± 1.60 × 10 ⁻³

* p < 0.05 significantly different from flux values for 25:75 EtOH/water vehicle.

Ethanol-Water Vehicles – Membrane Pretreatment Studies

The results of the pretreatment studies showed that the fluxes for the permeation of ibuprofen through silicone membrane pretreated with ethanol are not significantly higher than for the 25:75 and pure ethanol systems through untreated membrane (table 4). The 75:25 ethanol-water formulation has a slightly lower flux (~700 vs. ~900 $\mu\text{g}/\text{cm}^2/\text{h}$). As noted for the studies where the membrane was not pretreated, relatively low amounts of ethanol can significantly enhance permeation. The partition parameter values for the 100% ethanol vehicle is significantly higher than for the 25:75 and 75:25 vehicles (table 3), suggesting that the major effects of ethanol are to change the solubility properties of ibuprofen in the membrane thus enhancing flux.

Permeation Studies in Skin

Ibuprofen fluxes across skin are shown in table 5; the values were lower than those through silicone membrane (fig. 4). Diffusion across skin increased with the amount of ethanol in the formulation. These effects reflect the

ability of ethanol-water mixtures to modify the barrier function of the skin by a number of mechanisms which may include lipid fluidization [16, 17], lipid disordering [18], lipid extraction [19], and modulation of the lipid barrier and keratinized protein conformation at low ethanol concentrations [20].

The lower flux values for pure ethanol in human skin versus silicone reflect a direct effect of ethanol on the skin barrier. Ethanol and methanol alter the stratum corneum to a much greater degree than other alcohols [21] and the findings of the present study are in line with earlier observations [22]. In that work, ethanol flux through skin was a function of ethanol volume fraction in the donor solution and dehydration and shrinkage of the stratum corneum was seen after equilibration with pure ethanol. The decrease in oestradiol flux through human skin at ethanol-water ratios higher than 60:40 (v/v) observed before [15] has also been ascribed to ethanol dehydration effects on the stratum corneum.

No major changes in the diffusion or partition parameters were seen in the formulations containing ethanol (table 5). The partition parameters were all significantly

Table 5. Steady-state fluxes, thermodynamic and kinetic parameters, and permeability coefficients for the permeation of ibuprofen from saturated formulations containing ethanol and water through human epidermis (n = 5)

EtOH-water, %v/v	J, $\mu\text{g cm}^{-2} \text{h}^{-1}$	K·h, μm	D/h ² , h ⁻¹	k _p , cm h ⁻¹
0:100	24 ± 2.7	11,833 ± 503	0.23 ± 0.02	0.27 ± 0.03
25:75	48 ± 19*	3.81 ± 2.51	0.36 ± 0.24	$1.41 \times 10^{-4} \pm 5.76 \times 10^{-5}$
50:50	274 ± 33*	9.85 ± 2.14	0.61 ± 0.26	$5.40 \times 10^{-4} \pm 2.62 \times 10^{-4}$
75:25	293 ± 24*	23.30 ± 7.83	0.43 ± 0.30	$6.12 \times 10^{-4} \pm 4.68 \times 10^{-5}$
100:0	60 ± 24*	24.1 ± 2.8	0.07 ± 0.03	$1.76 \times 10^{-4} \pm 1.01 \times 10^{-4}$

* Significantly different from flux values for aqueous vehicle; p < 0.05.

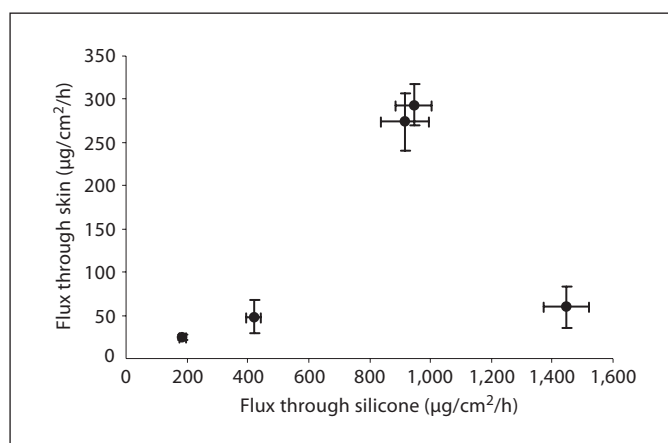


Fig. 4. Steady-state ibuprofen fluxes, from saturated formulations containing ethanol and water, through silicone membranes and human skin.

less than the Kh value for the aqueous ibuprofen solution. These effects are similar to those observed for oestradiol [15] and in other published work [16, 23].

Conclusions

We have previously evaluated the skin penetration of topical formulations of ibuprofen and we have shown that the composition of the vehicles can have a significant impact on the percutaneous penetration of the drug [2, 24–27]. A typical cream formulation may contain up to 10% of a cosolvent which can alter the pK_a of a weak acid or a weak base and change thereby its solubility and partitioning behaviour. As these changes will influence percutaneous

penetration, the present study was designed to explore the modulation of ibuprofen flux across silicone membrane and skin. In the lipophilic environment of the stratum corneum, an ionized species is expected to permeate more slowly than a unionized moiety. From the pK_a and log D measurements, increasing amounts of ethanol increased the pK_a and the proportion of unionized ibuprofen, enhancing the partitioning of the compound. The presence of ibuprofen dimers at higher ethanol concentrations may also influence partitioning behaviour. The actual effects of ethanol on partitioning and diffusion are difficult to deconvolve from conventional permeation studies as can be seen from the present work. For both skin and silicone, flux increased with increasing amounts of ethanol; however, pure ethanol solutions did not enhance the skin permeation of ibuprofen. Both silicone and skin permeation data suggested that the influence of ethanol is primarily due to enhanced solubility of ibuprofen in the membrane.

The addition of the cosolvent clearly has a significant effect upon the physicochemical properties of ibuprofen in the systems studied. The vehicles used in the present study were simple binary ethanol-water systems, whereas conventional topical formulations will generally contain a far greater number of excipients. Future studies will examine more complex systems and will also investigate how the partition and diffusion parameters change with the specific vehicle.

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References

- Hadgraft J, Valenta C: pH, pK_a and dermal delivery. *Int J Pharm* 2000;200:243–247.
- Hadgraft J, Whitefield M, Rosher PH: Skin penetration of topical formulations of ibuprofen 5%: an in vitro comparative study. *Skin Pharmacol Appl Skin Physiol* 2003;16:137–142.
- Wade A: *Pharmaceutical Handbook*, ed 19. London, The Pharmaceutical Press, 1980.
- Avdeef A, Comer JEA, Thomson SJ: pH-metric log P. 3. Glass electrode calibration in methanol-water, applied to pK_a determination of water-insoluble substances. *Anal Chem* 1993;65:42–49.
- Avdeef A, Box KJ, Comer JE, Gilges M, Hadley M, Hibbert C, Patterson W, Tam KY: pH-metric log P 11. pK_a determination of water-insoluble drugs in organic solvent-water mixtures. *J Pharm Biomed Anal* 1999;20:631–641.
- Yasuda M: Dissociation constants of some carboxylic acids in some mixed aqueous solvents. *Bull Chem Soc Jpn* 1959;32:429–432.
- Shedlovsky T: The behaviour of carboxylic acids in mixed solvents; in Pesce B (ed): *Electrolytes*. New York, Pergamon Press, 1962, pp 146–151.
- Dias M, Hadgraft J, Lane ME: Influence of membrane solvent solute interactions on solute permeation in model membranes. *Int J Pharm* 2007;336:108–114.
- Kligman AM, Christophers E: Preparation of isolated sheets of human stratum corneum. *Arch Dermatol* 1963;88:70–73.
- Diez-Sales O, Watkinson M, Herraiz-Dominguez M, Javaloyes C, Hadgraft J: A mechanistic investigation of the in vitro human skin permeation enhancing effect of Azone®. *Int J Pharm* 1996;129:33–40.
- Albert A, Serjeant EP: *The Determination of Ionization Constants – A Laboratory Manual*, ed 3. New York, Chapman & Hall, 1984.
- Watkinson AC, Brain KR, Walters KA: The penetration of ibuprofen through human skin in vitro: vehicle, enhancer and pH effects; in Brain KR, James V, Walters KA (eds): *Prediction of Percutaneous Penetration 3B*. Cardiff, STS Publishing, 1993, pp 335–341.
- Hoy KL: New values of the solubility parameters from vapour pressure data. *J Paint Technol* 1970;42:76–118.
- Chan KLA, Kazarian SG: High-throughput study of poly(ethylene glycol)/ibuprofen formulations under controlled environment using FTIR imaging. *J Comb Chem* 2006;8:26–31.
- Megrab NA, Williams AC, Barry BW: Oestradiol permeation across human skin, silastic and snake skin membranes: the effects of ethanol/water cosolvent systems. *Int J Pharm* 1995;116:101–112.
- Ghanem AH, Mahmoud H, Higuchi WI, Rohr UD, Boradia S, Liu P, Fox JL, Good WR: The effects of ethanol on the transport of β -oestradiol and other permeants in hairless mouse skin. II. A new quantitative approach. *J Control Release* 1987;6:75–83.
- Higuchi WI, Rohr UD, Burton SA, Liu P, Fox JL, Ghanem AH, Mahmoud H, Borsadia S, Good WR: Effect of ethanol on the transport of β -estradiol in hairless mouse skin. Comparison of experimental data with a new theoretical model; in Lee PI, Good WR (eds): *Controlled Release Technology. Pharmaceutical Applications* (ACS Symp. Ser. 348). Washington, American Chemical Society, 1987, pp 232–240.
- Rowe ES: Thermodynamic reversibility of phase transitions. Specific effects of alcohols on phosphatidylcholines. *Biochim Biophys Acta* 1985;813:321–330.
- Bommannan D, Potts RO, Guy RH: Examination of the effect of ethanol on human stratum corneum in vivo using infrared spectroscopy. *J Control Release* 1991;16:299–304.
- Kurihara-Bergstrom T, Knutson K, DeNoble J, Goates CY: Percutaneous absorption enhancement of ionic molecules by ethanol-water systems in human skin. *Pharm Res* 1990;7:762–766.
- Scheuplein RJ, Blank IH: Mechanism of percutaneous absorption. IV. Penetration of nonelectrolytes (alcohols) from aqueous solutions and from pure liquids. *J Invest Dermatol* 1973;60:286–296.
- Berner B, Juang RH, Mazzenga GC: Ethanol and water sorption into stratum corneum and model systems. *J Pharm Sci* 1989;78:472–476.
- Pershing LK, Lambert LD, Knutson K: Mechanism of ethanol enhanced estradiol permeation across human skin in vivo. *Pharm Res* 1990;7:170–175.
- Herkenne C, Naik A, Kalia YN, Hadgraft J, Guy RH: Pig ear skin ex vivo as a model for in vivo dermatopharmacokinetic studies in man. *Pharm Res* 2006;23:1850–1856.
- Herkenne C, Naik A, Kalia YN, Hadgraft J, Guy RH: Ibuprofen transport into and through skin from topical formulations: in vitro-in vivo comparison. *J Invest Dermatol* 2007;127:135–142.
- Herkenne C, Naik A, Kalia YN, Hadgraft J, Guy RH: Dermatopharmacokinetic prediction of topical drug bioavailability in vivo. *J Invest Dermatol* 2007;127:887–894.
- Herkenne C, Naik A, Kalia YN, Hadgraft J, Guy RH: Effect of propylene glycol on ibuprofen absorption into human skin in vivo. *J Pharm Sci* 2008;97:185–197.